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Semester: 8
Paper Name: Antennas & Propagation
Paper Code: PE-EC801A

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Introduction

- An antenna is a device that transmits and receives electromagnetic waves.
- It acts as a bridge between electrical circuits and free space by converting electrical energy into radio waves and radio waves back into electrical signals.
- Antennas are essential components in modern communication systems such as mobile networks, television broadcasting, Wi-Fi, radar, and satellite communication.
- The performance of any wireless system mainly depends on the type and design of the antenna used.
- Important parameters of antennas include gain, directivity, radiation pattern, bandwidth, and efficiency.
- Among various types of antennas, Array Antennas and Reflector Antennas are widely used for high gain and directional communication.

Array Antenna

- An Array Antenna is formed by combining two or more individual antenna elements.
- These elements are arranged in a specific geometrical pattern and are fed with controlled amplitude and phase.
- By adjusting the spacing and phase difference, the radiation pattern can be controlled.
- It increases signal strength and improves directionality.
- Types: Broadside Array, End-fire Array & Phased Array

Advantages & Applications of Array Antenna

Advantages:

- High directivity and gain
- Beam steering without physically moving the antenna (in phased arrays)
- Reduced interference and improved signal quality
- Flexible radiation pattern control

Applications:

- Radar systems
- 5G and wireless communication
- Satellite communication
- Military and defense systems

Reflector Antenna

- A Reflector Antenna uses a reflecting surface to direct electromagnetic waves in a particular direction.
- It consists of a feed antenna placed in front of a metallic reflecting surface.
- The reflector collects and focuses the signals to produce a strong directional beam.
- The most common reflector shape is parabolic because it focuses signals at one focal point.

Types, Advantages & Applications of Reflector Antenna

Types:

- Plane Reflector
- Corner Reflector
- Parabolic Reflector

Applications:

- Satellite dishes
- TV broadcasting
- Radio telescopes
- Radar systems

Advantages:

- Very high gain
- Narrow beamwidth
- Suitable for long-distance communication
- Simple construction

Comparison between Array & Reflector Antenna

Array Antenna

- Multiple elements
- Electronic (phased array) beam control
- High gain
- Costly

Reflector Antenna

- Single feed + reflector
- Mechanically adjusted beam control
- Very high gain
- Moderate costly

Conclusion

- Both Array Antennas and Reflector Antennas are important in modern wireless communication systems.
- Array antennas provide flexibility, electronic beam steering, and better control over radiation patterns, making them suitable for advanced technologies like 5G and smart radar systems.
- Reflector antennas provide extremely high gain and are ideal for long-distance communication such as satellite links and space communication.
- The selection of antenna type depends on application requirements like range, cost, complexity, and beam control.
- With the rapid growth of wireless technology, both these antennas continue to play a major role in communication, defense, broadcasting, and space research.

THANK YOU